

CSC3044S
Artificial Intelligence: AI Systems
Introduction II

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Recap: Rational Agents

- **Rational:** Maximally achieving pre-defined goals.
- Goals are expressed in terms of the **utility** of outcomes.
- Being rational means **maximising your expected utility**.

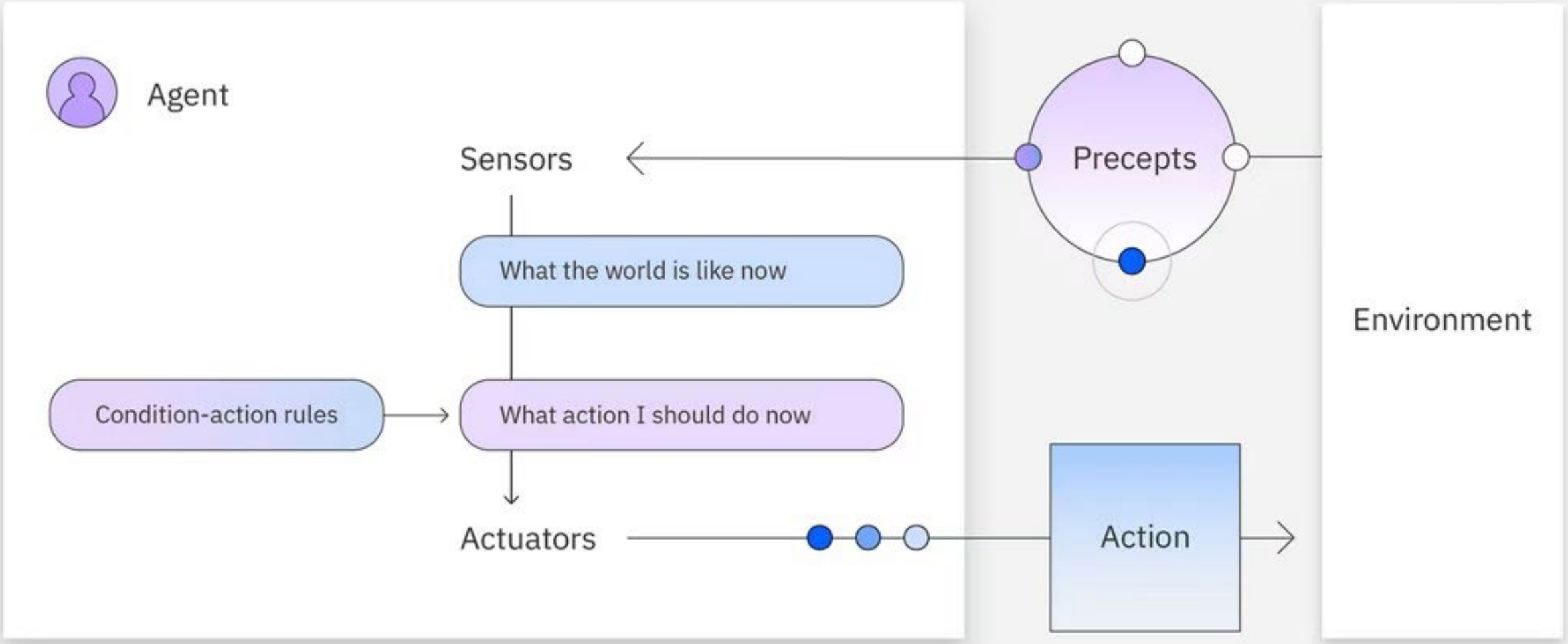
Recap: Types of (Rational) Agents

- **Reactive** (Reflexive) agents – Simple and Model-based.
- **Deliberative** (Goal-based) agents.
  - **Learning** (RL, ANN, LLM,…) agents.

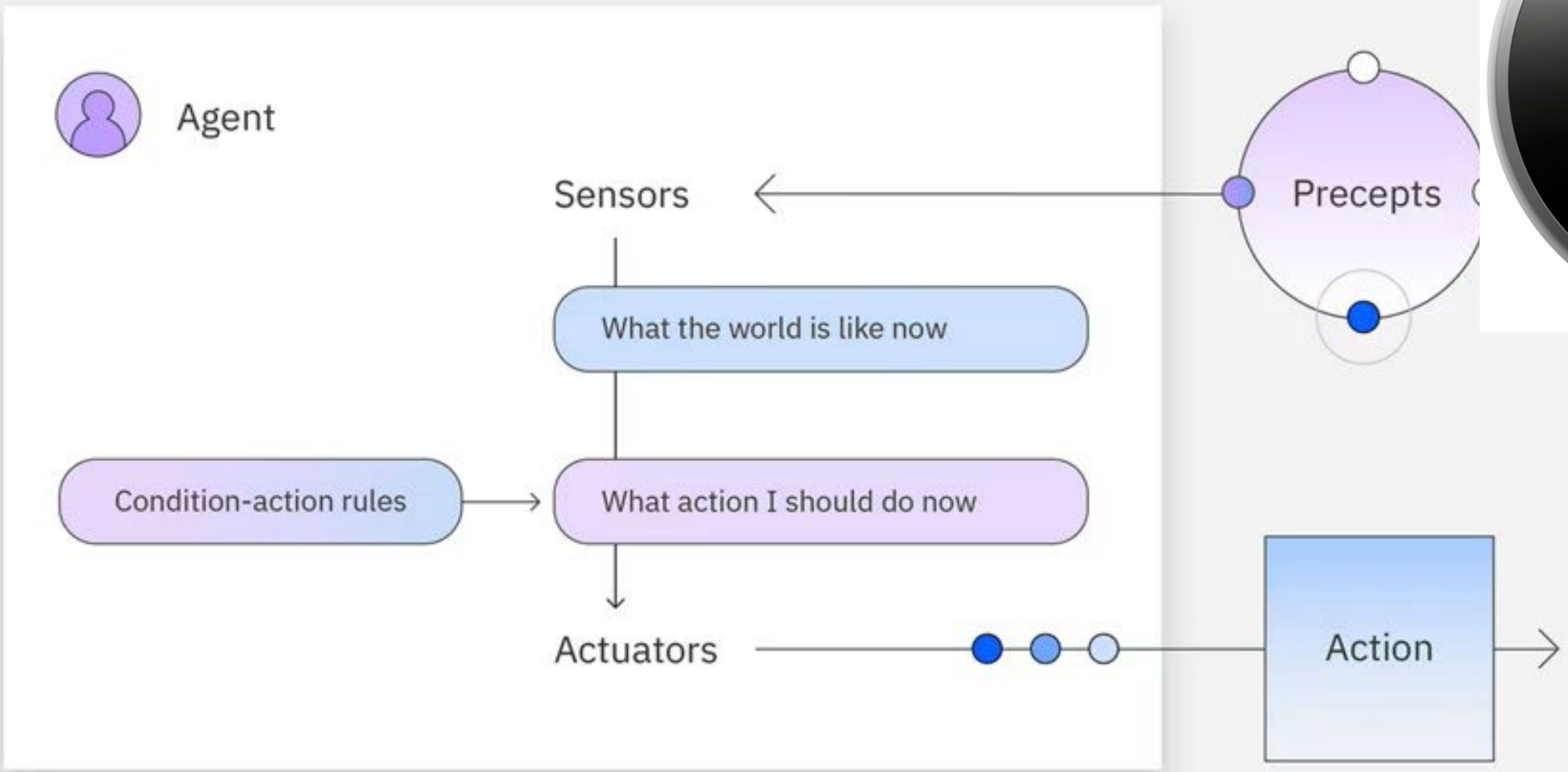
Reactive Agents: Stimulus–Response

- Agents just react to the current (sensed) environment state.
- Agents sense their environment, react to any new information.
- Behaviour is dictated by immediate sensory input and rules mapping inputs → output S.
- Agents lack any memory, internal state, explicit description of the environment or goals.

Reactive Agent – Example?



Reactive Agent





Reactive Agent Architectures

- Agent behaviour continually maps (sensed) inputs to actions.

- Reactive behaviour: action: $S \rightarrow A$
  - Where, *S* denotes the states of the environment, and A the primitive actions the agent is capable of doing.

- Example: action(s) =
  - *Heater on*, if temperature too low
  - *Heater off*, otherwise

Reactive Agents – Problems?

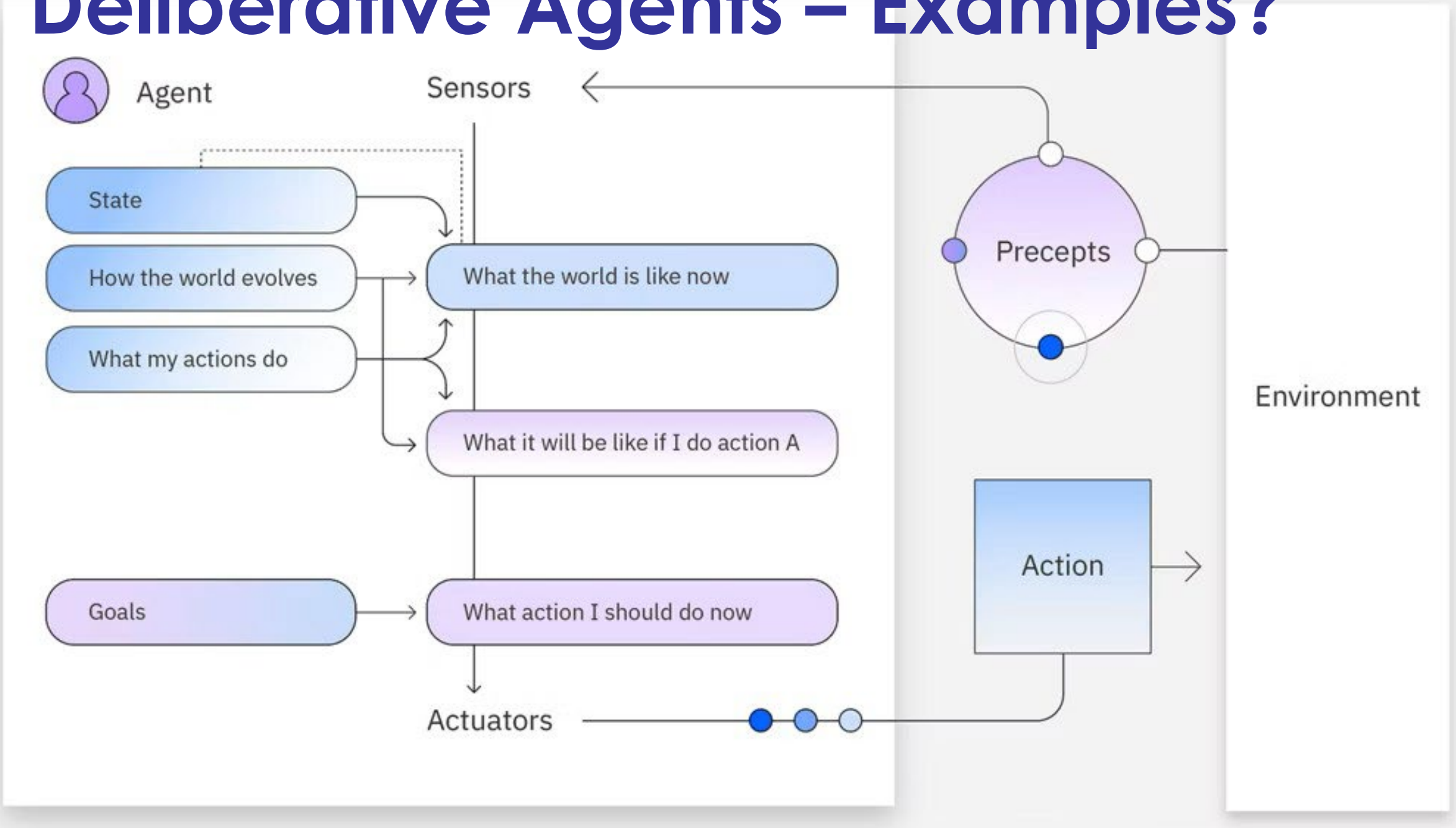
Reactive Agents – Problems?

- Agents are “handcrafted” – Development takes a lot “tuning”
- Agents can be used only for its original purpose.
- Impossible to build large systems?
- How to learn?

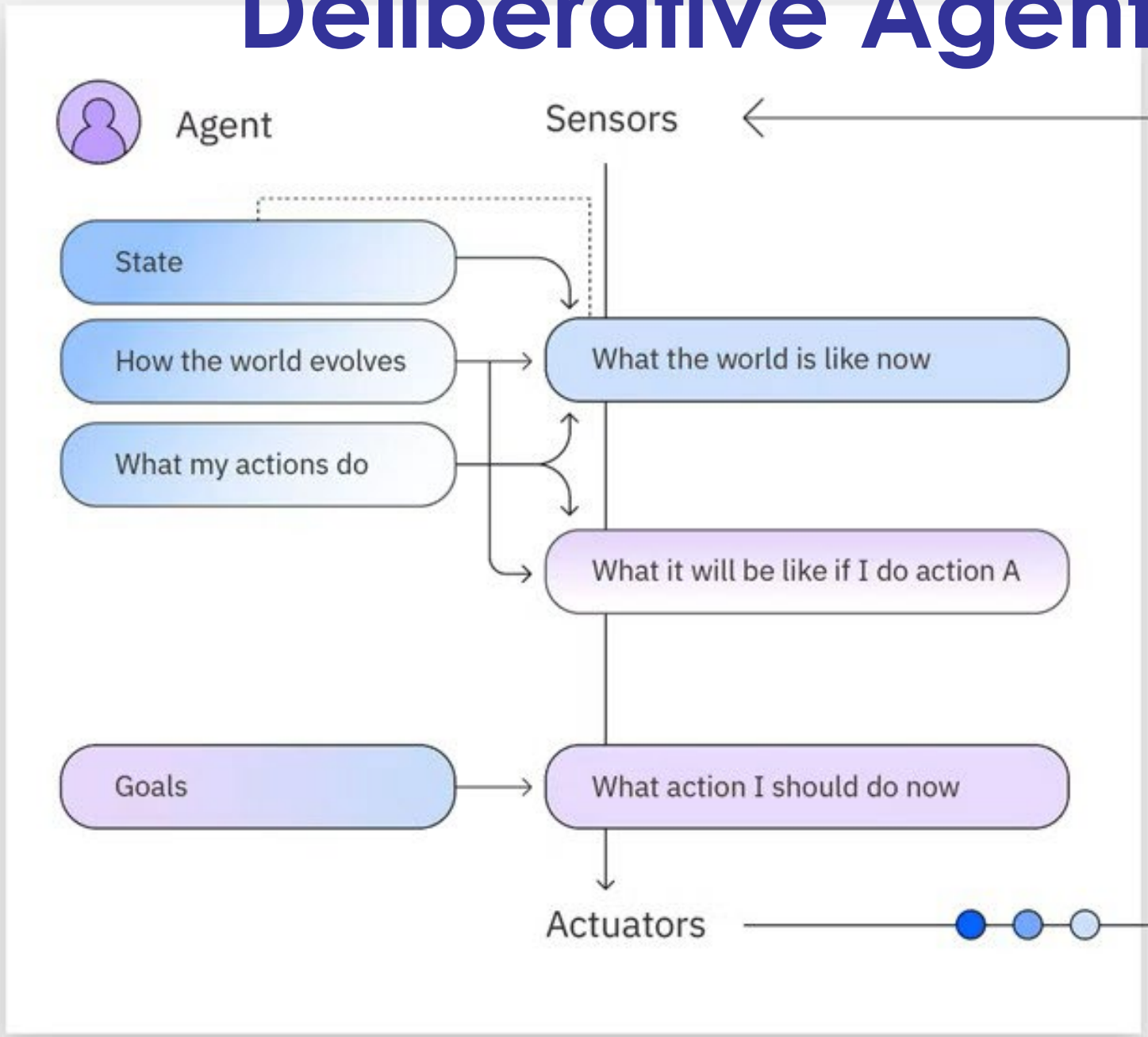
Deliberative Agents: Sense-plan-act

- Agents sense their environment, generate a plan – act on plan.
- Agent behaviour dictated by:
  - (Sensed) environment state,
  - (Agent) internal state,
  - Plan that maps inputs → outputs (e.g., logical reasoning, symbolic manipulation, rule-set, ANN, …).
- Deliberative agents usually have memory, internal state, explicit description of the environment and goals.

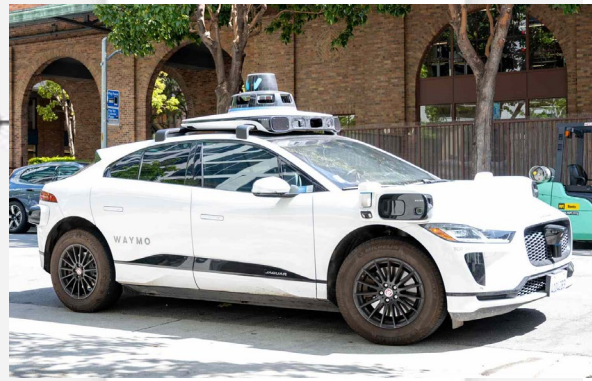
Deliberative Agents – Examples?



Deliberative Agents







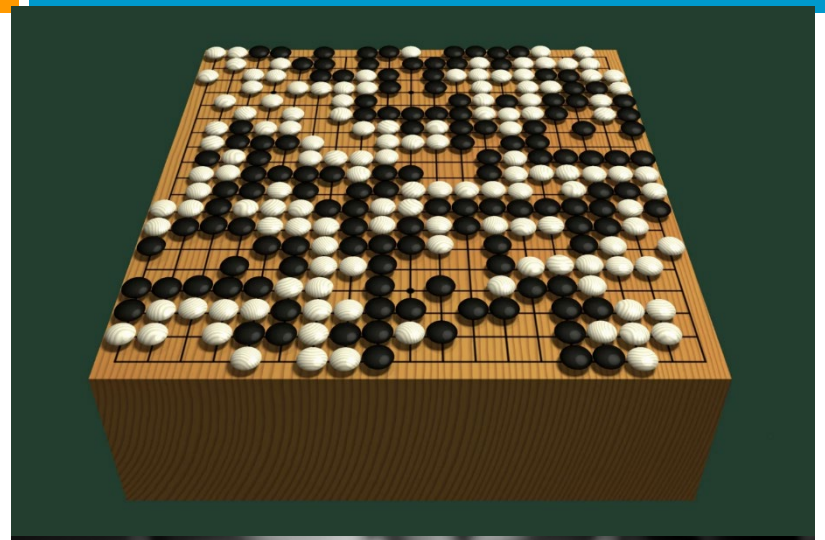
Claude BY ANTHROP\C



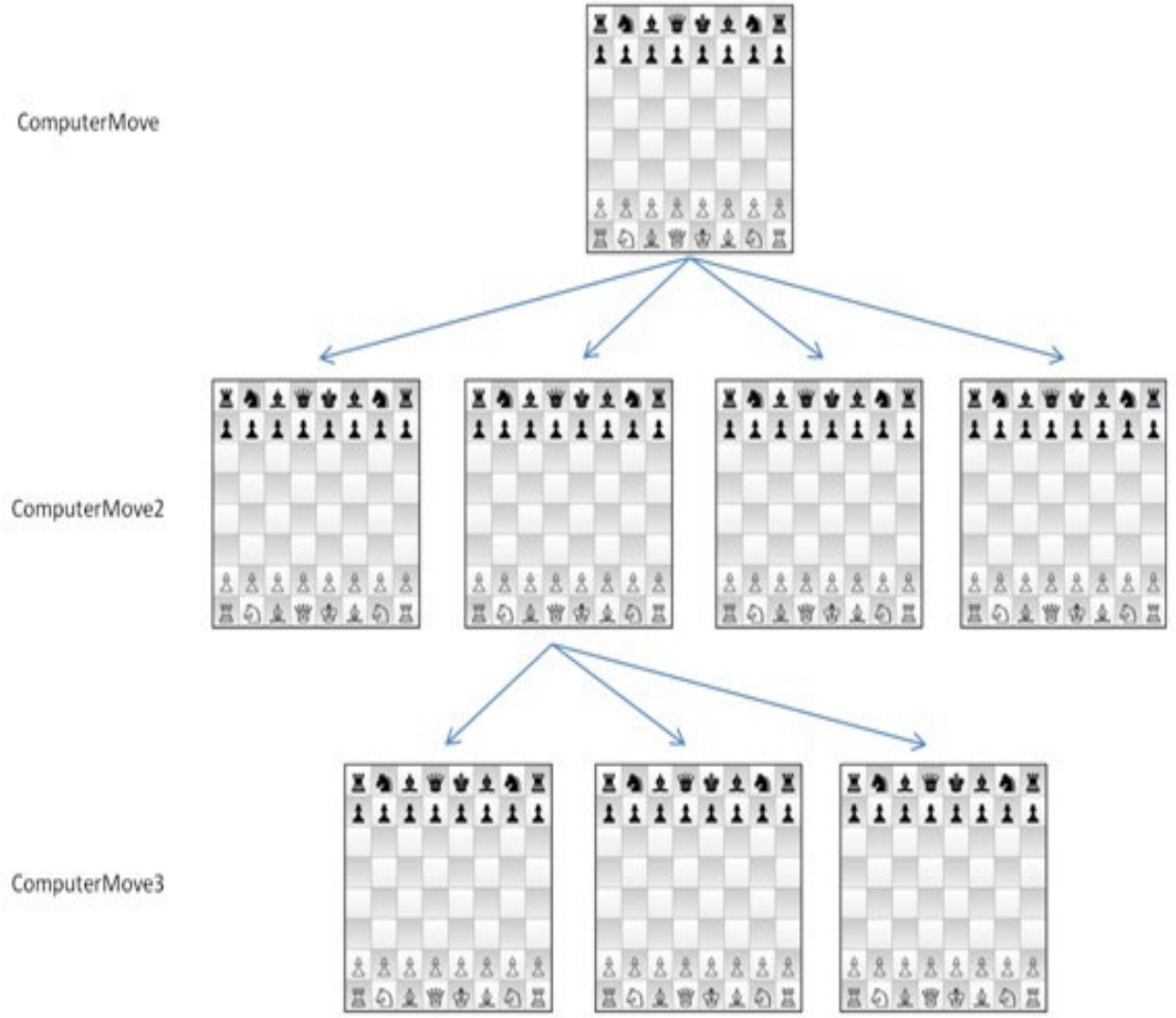
How do Deliberative Agents work?

- **(Most)** Deliberative agents use Symbolic AI algorithms.
  - Contain an explicitly represented, symbolic model of the world.
  - Makes decisions (actions to perform) via symbolic reasoning (manipulating symbols, e.g., logic, rule-sets, decision-trees).
- An example of a “classic” (Symbolic AI) deliberative agent?

Game-Playing Agents: Top-down Symbolic Algorithms







Deliberative Agents – Problems?

Deliberative Agents – Problems?

- **Transduction problem:** Impractical to translate needed information into symbolic representations – especially if the environment is changing (dynamic).
- **Representation problem: … ?**
- Limited adaptivity.
- Do not scale to real-world scenarios.

The (AI) Representation Problem

- **Knowledge Representation:** How to encode facts, rules, causality, time in (agent) models? (e.g., Bayes rule vs neural networks).

- **Problem Representation:** How to define a task (environment) that a model must operate in (e.g., Chess vs autonomous vehicle).

- **Data Representation:** How to translate unstructured data (e.g., captured text, images, audio) into useful data-structures (e.g., high-dimensional vectors as neural network input layers).

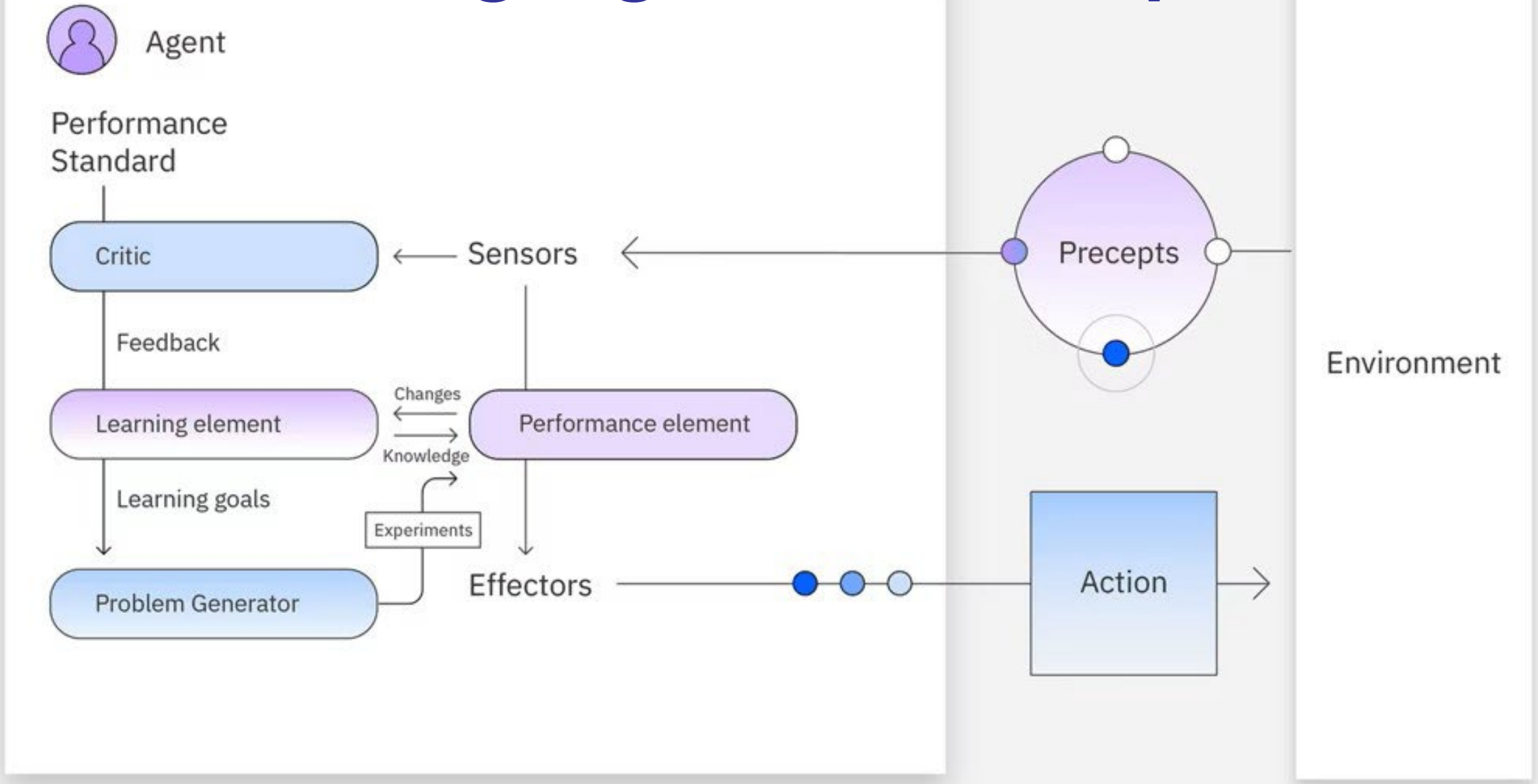
The Representation Problem: Generally

- ***Main issue: Complexity of symbol manipulation algorithms***.

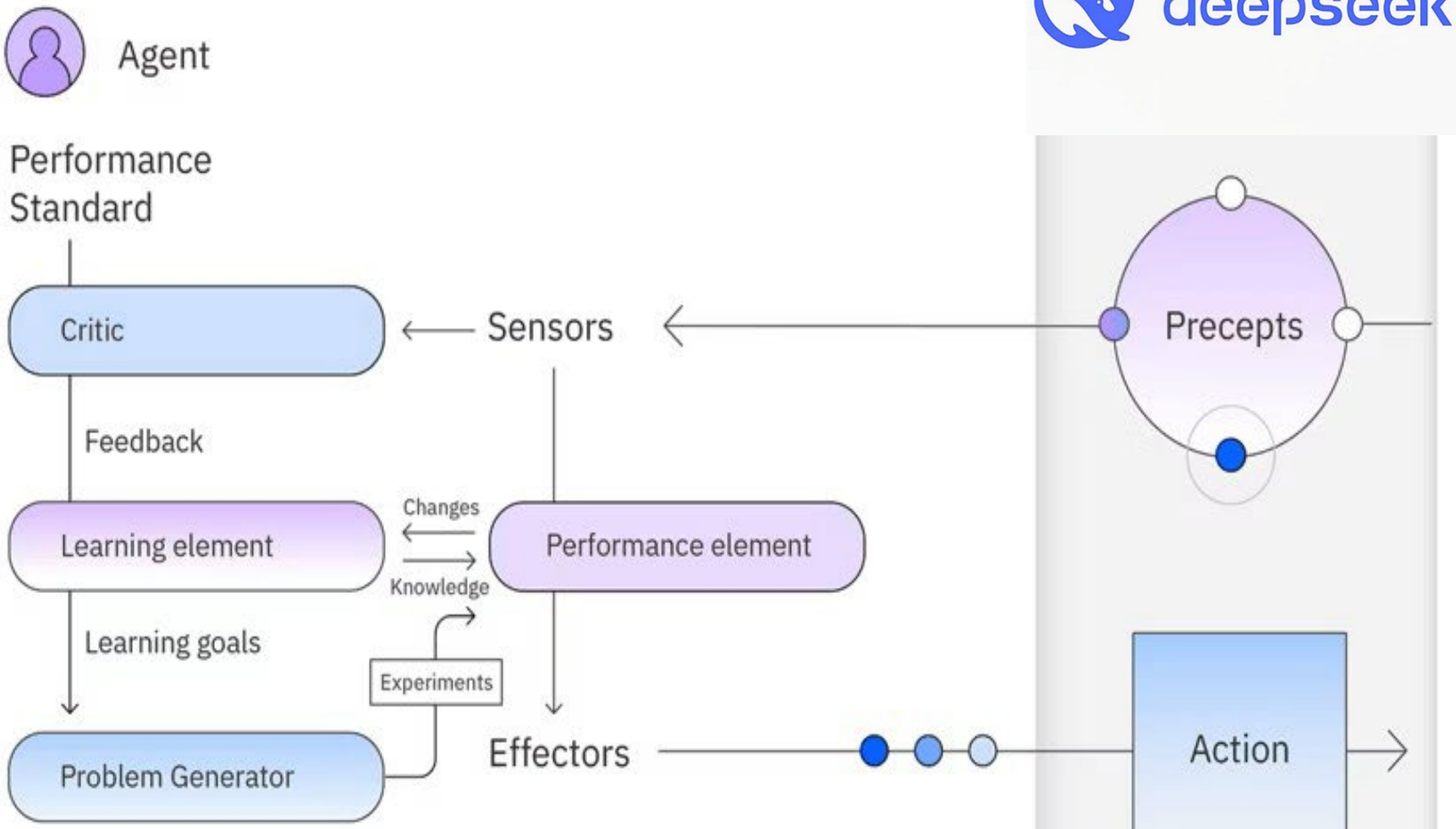
- Most (search-based) symbol manipulation algorithms are highly intractable and limited (e.g., game-playing algorithms).

- Hard to find compact representations for most problems – thus the rise of other agent architectures (e.g., neuro-AI based).

Learning Agents – Examples?

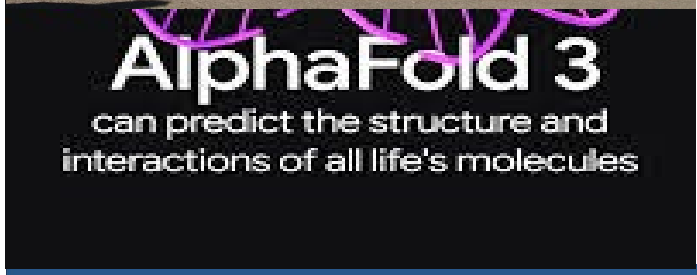


Learning Agents



deepseek

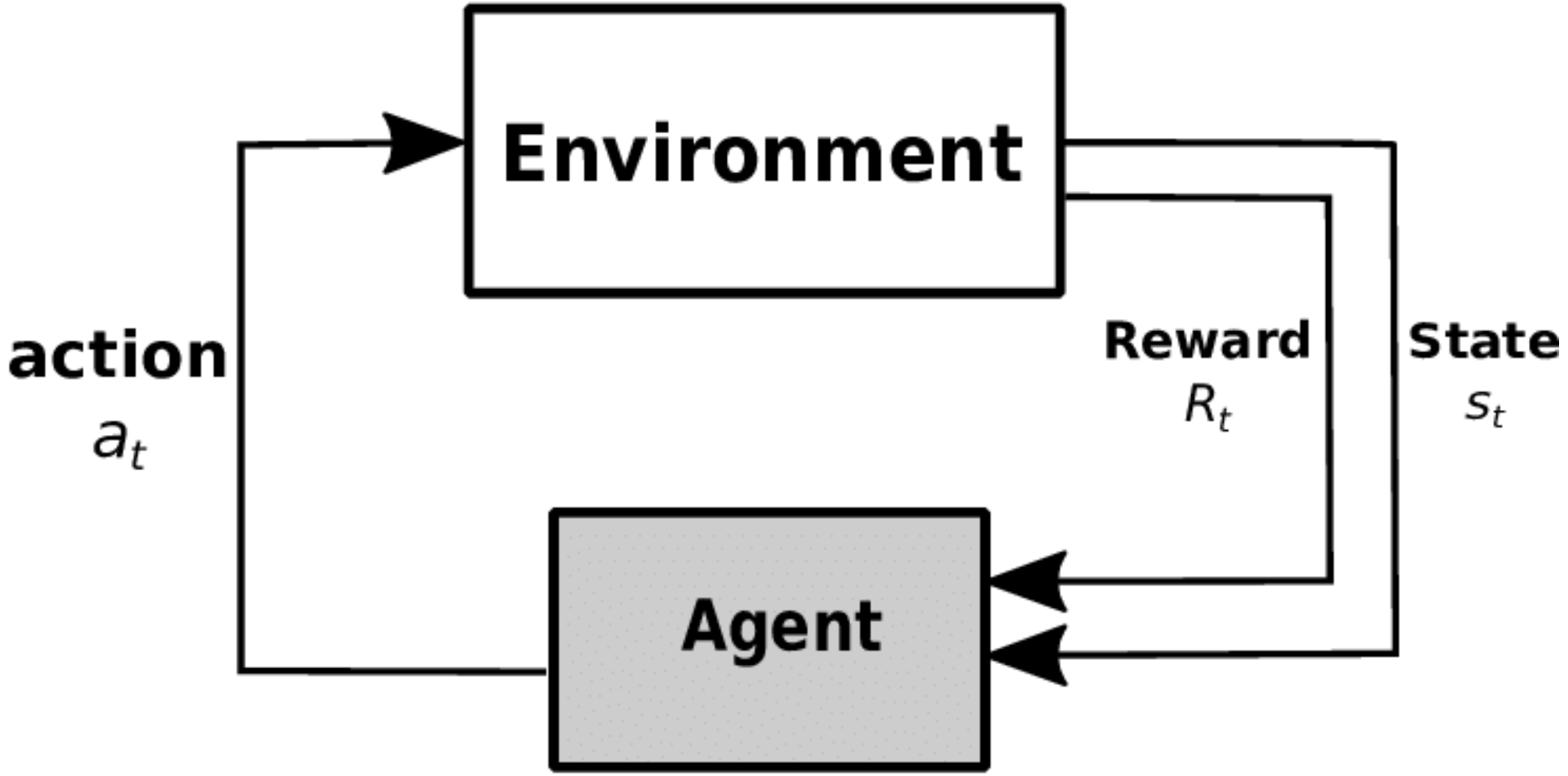




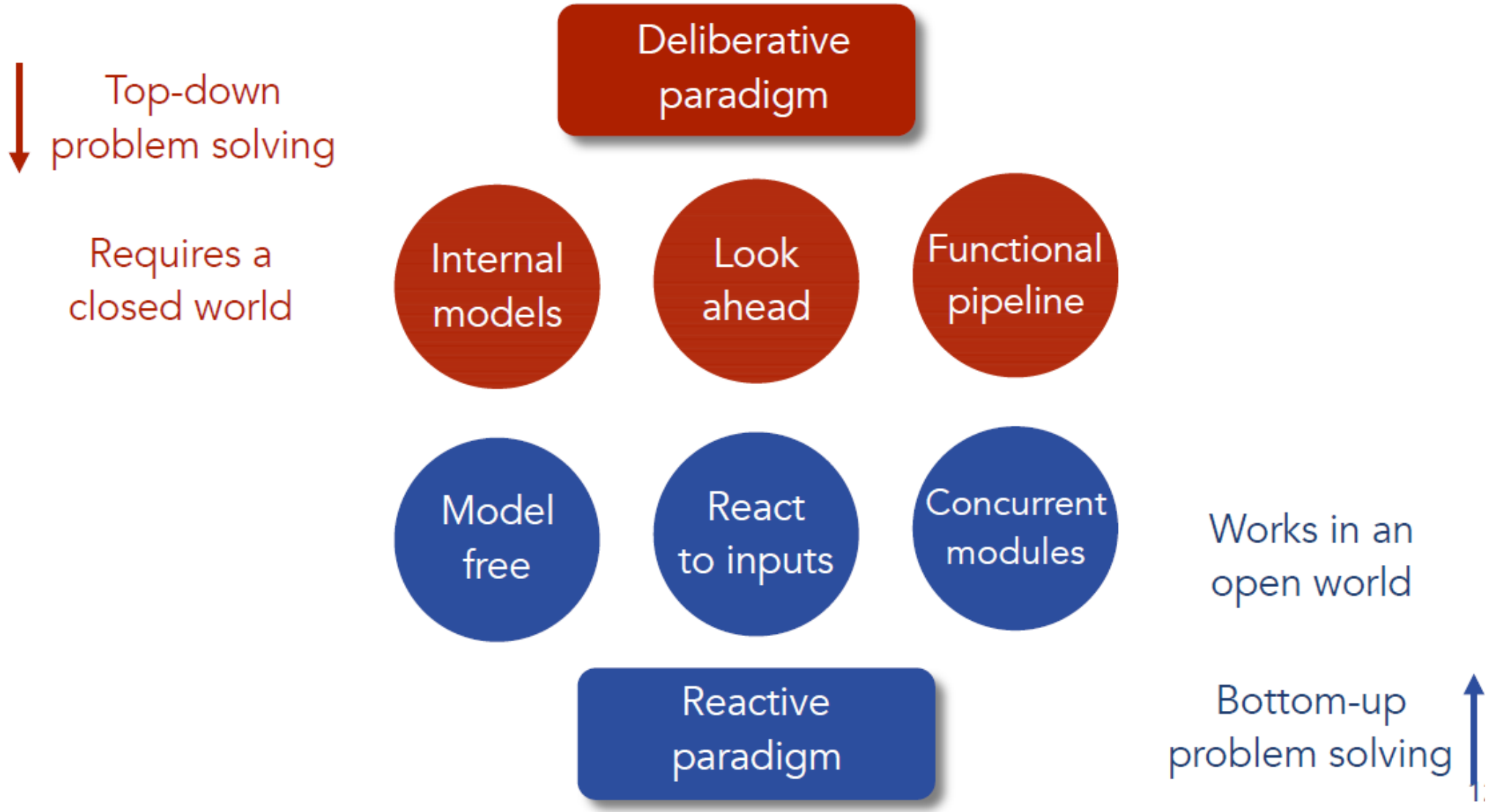




Reinforcement Learning Agent



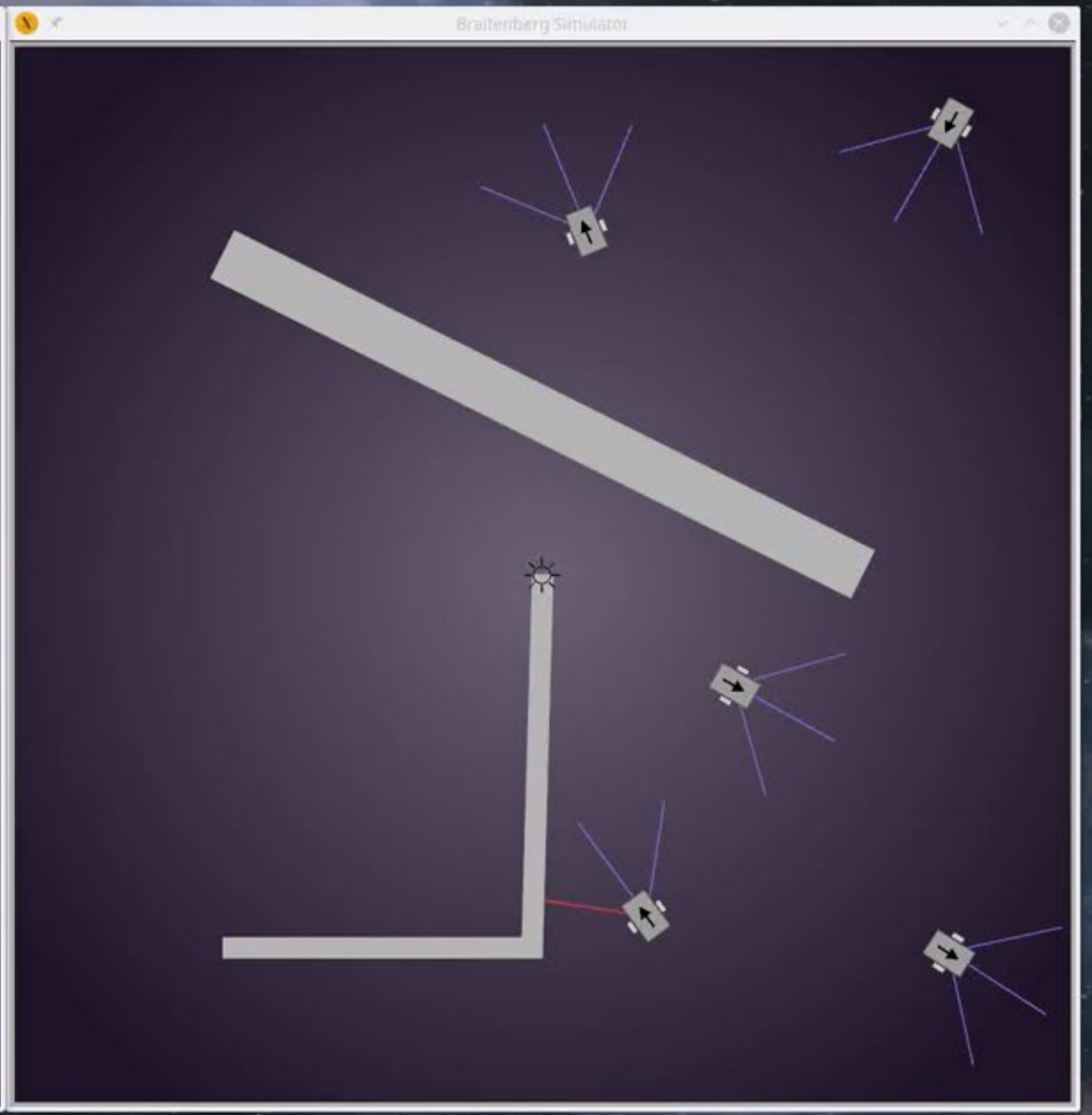
Reactive vs Deliberative Paradigms



Braitenberg's Vehicles:

Reactive
OR
Deliberative (No learning)
OR
Deliberative (Learning)?

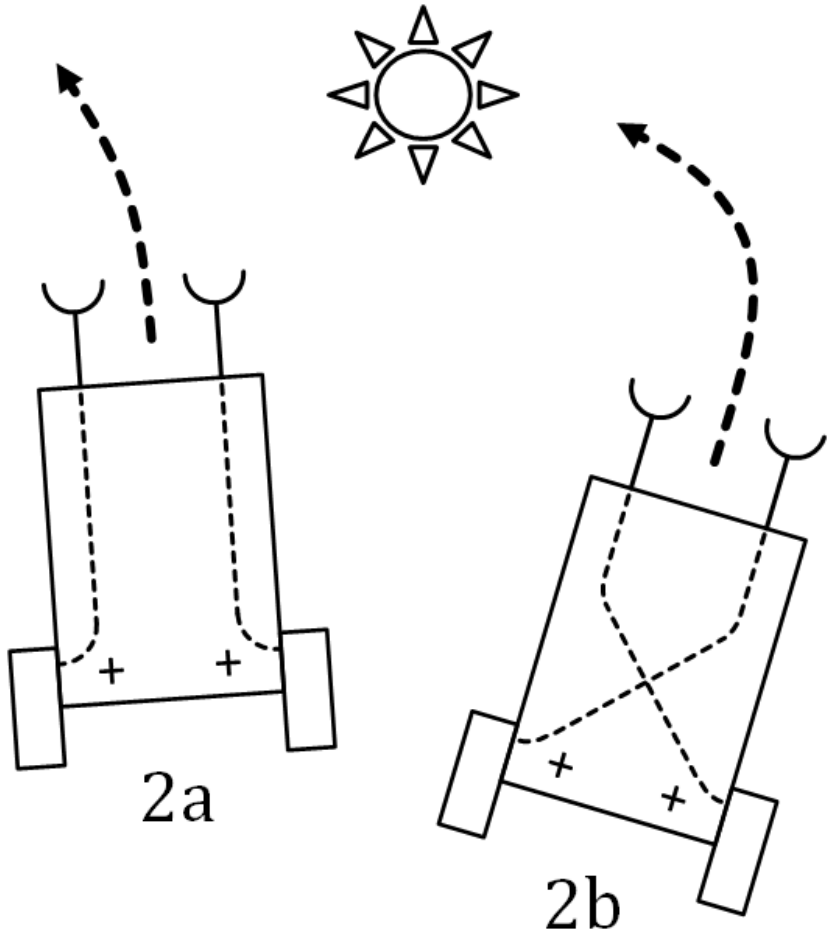

Braitenberg Simulator


Braitenberg Simulator

Braitenberg's Vehicles: Reactive, Deliberative (No learning) or Deliberative (Learning)?

Avoid Light (2a):

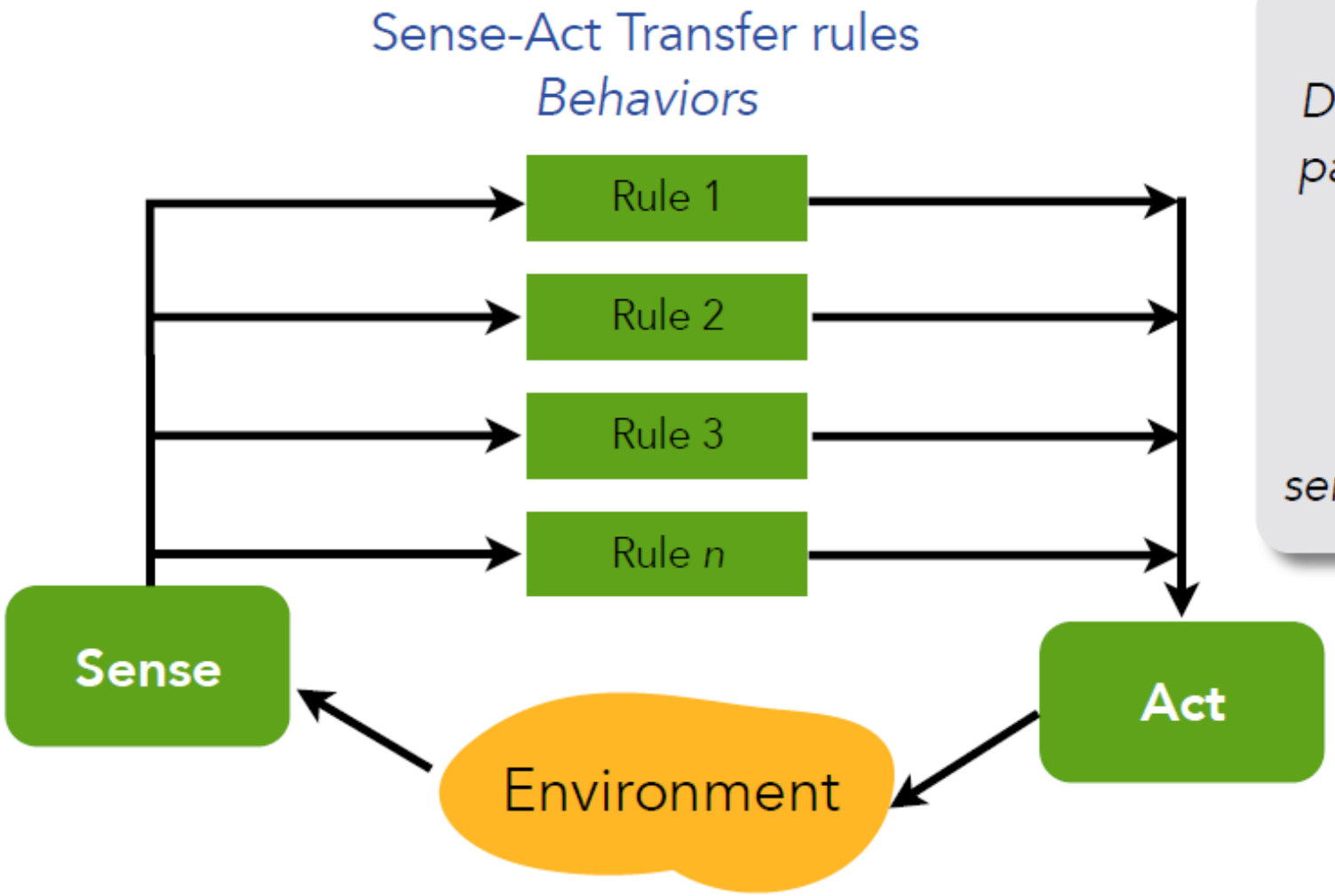
IF More light to right:
→ right wheel turns faster
→ turns towards the left, away from the light.



Goto Light (2b):

IF More light to left:
→ right wheel turns faster
→ turns towards the left, to be closer to the light.

Reactive (Sense-Act) Agent Controllers

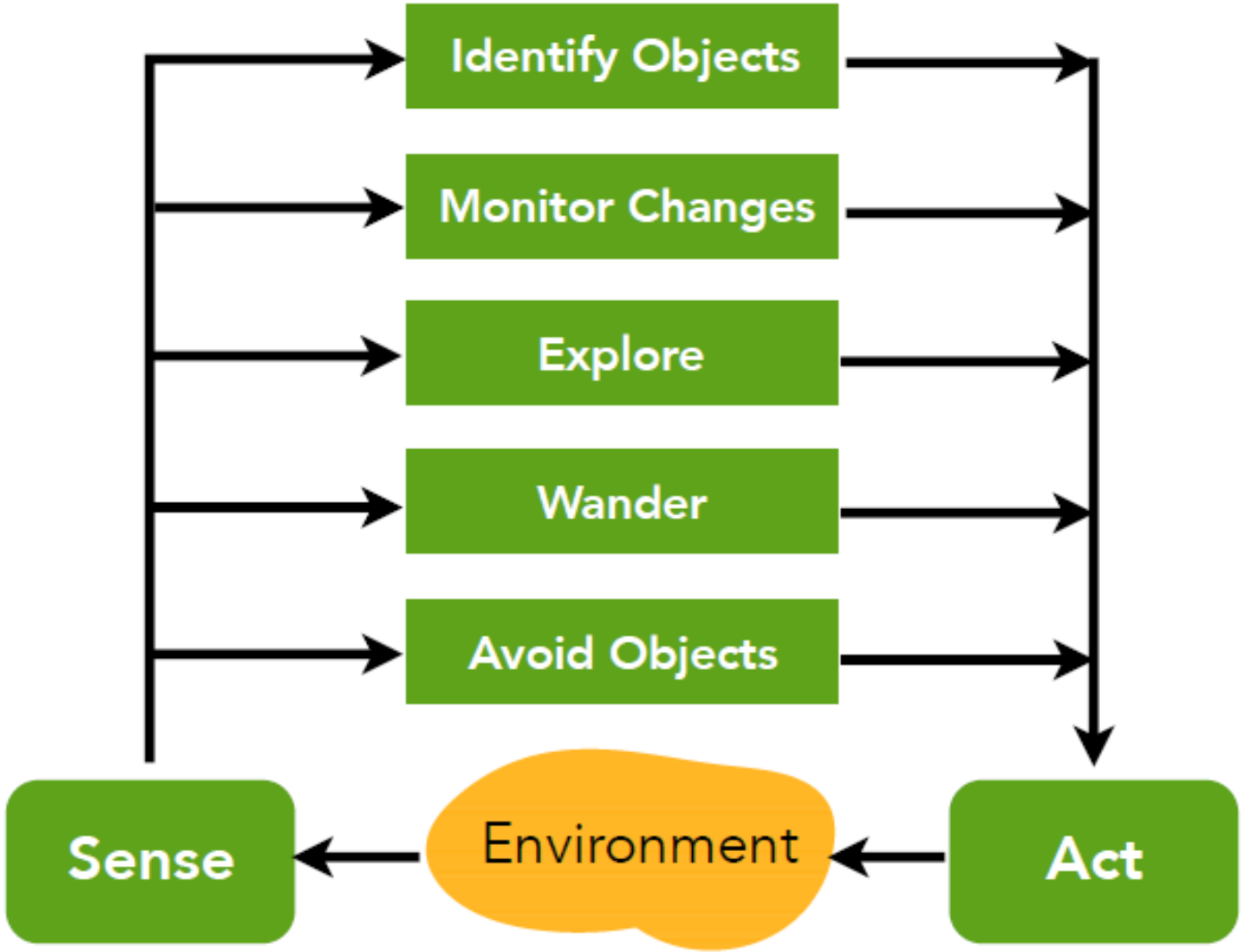


Ethological view (***Behavior***):
Direct mapping of sensory inputs to a pattern of motor actions that are then used to achieve a task

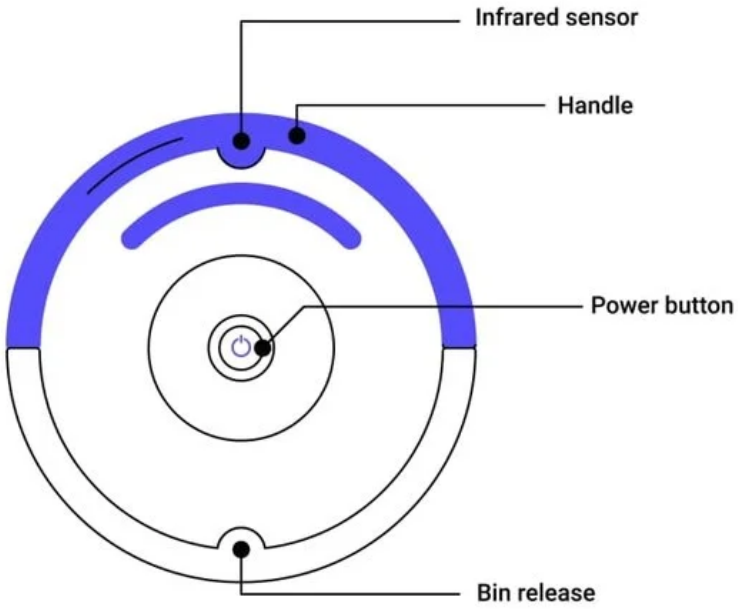
Mathematical view (***Function***):
A transfer function, transforming sensory inputs into actuator commands

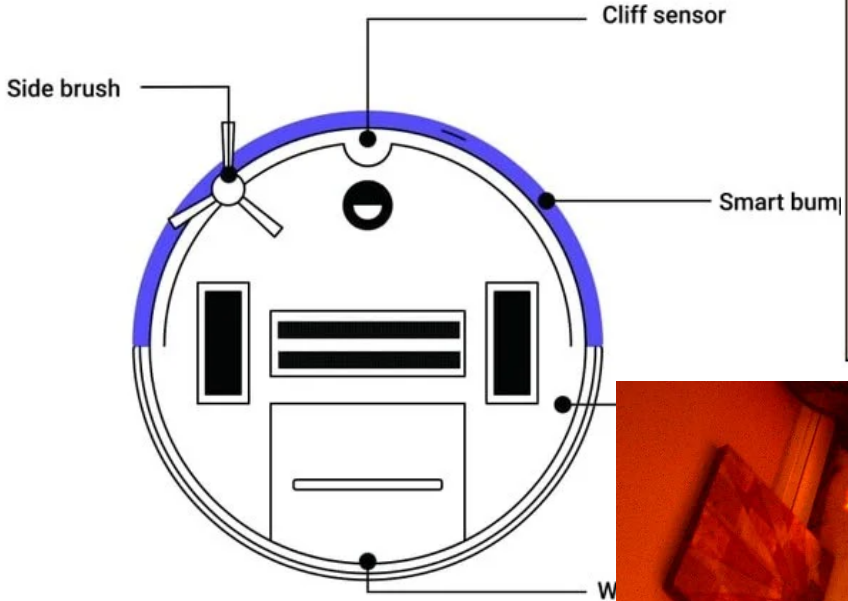
Concurrent mode
vs.
Sequential mode

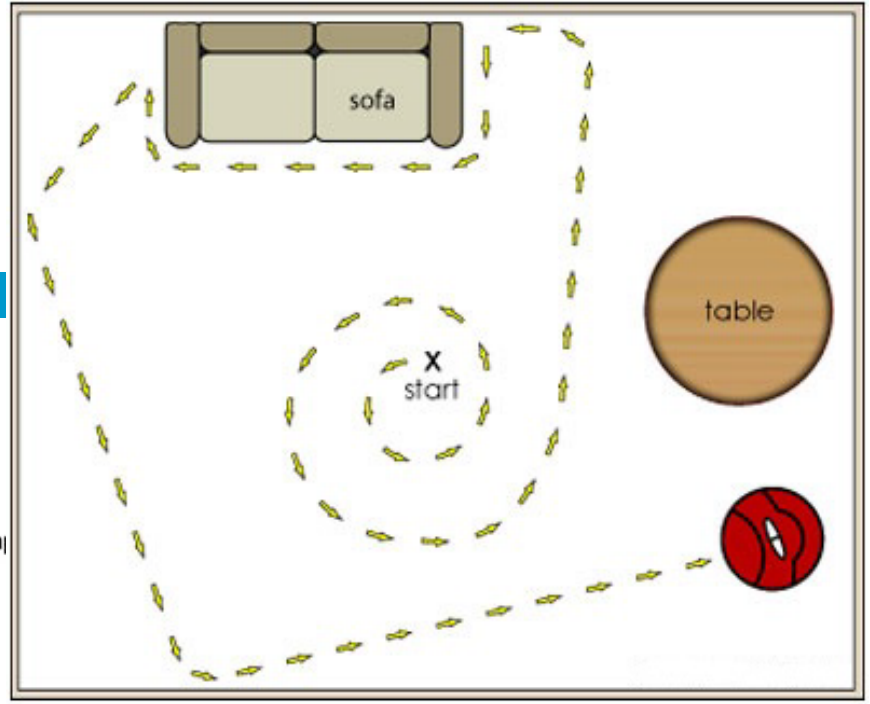
Example Sense-Act Controller



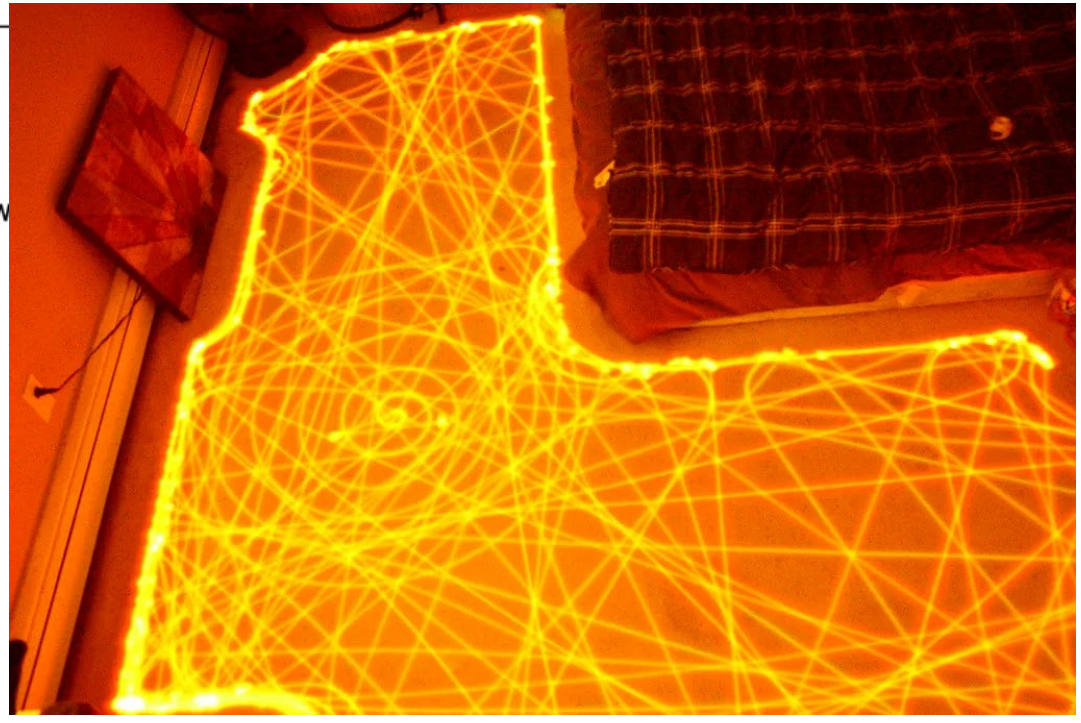
Roomba Vacuum Cleaner











What about some Deliberative Agent Applications?



OS Agents: Are there any AI Operating Systems?

Are there an AI Operating Systems?

There are currently no fully autonomous **AI operating systems (AI OS)** in the sense of a general-purpose OS entirely powered or managed by AI. However, several **AI-enhanced operating systems** and **AI-native platforms** do exist, and efforts are ongoing to make operating systems more intelligent. Here's a breakdown of what's out there:

✅ AI-Enhanced Operating Systems

These are traditional OSes (like Windows, macOS, Linux, Android) that have integrated AI features:

- **Windows 11 Copilot**: Microsoft integrates AI (via Copilot and Bing Chat, based on OpenAI's models) to enhance user interaction and productivity directly in the OS.
- **macOS / iOS**: Apple uses AI for features like Siri, photo recognition, predictive typing, and system optimizations.
- **Android**: Google integrates AI through Google Assistant, app suggestions, adaptive battery, and more.

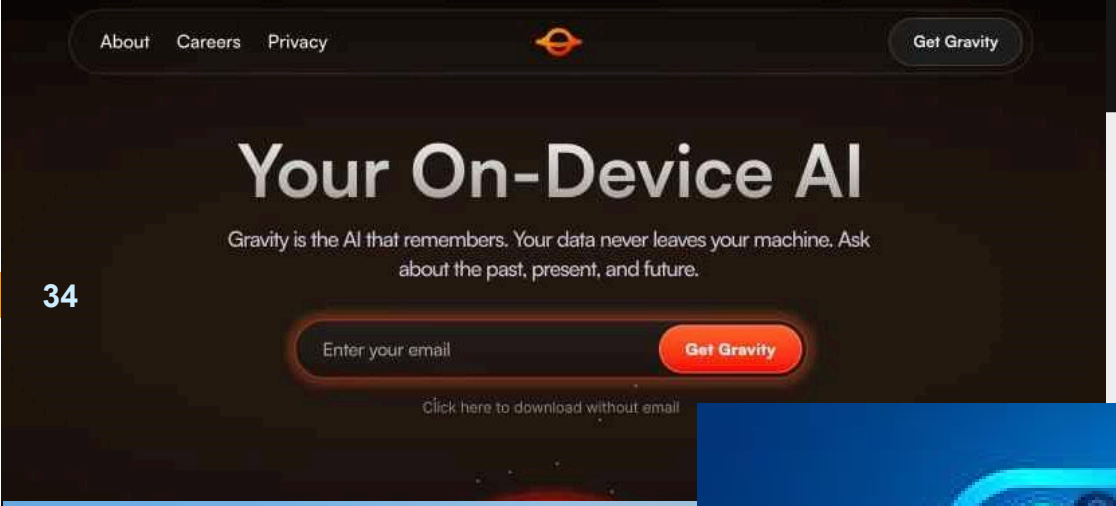
Are there any AI Operating Systems?

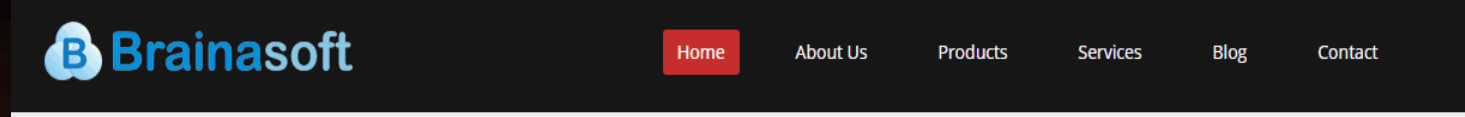
🧠 Vision for the Future

A true **AI Operating System** would involve:

- **Autonomous resource management** (RAM, CPU, tasks)
- **Conversational interface as the shell**
- **AI-driven scheduling and optimization**
- **Dynamic UI personalization**
- **Integration with agents that learn and evolve**

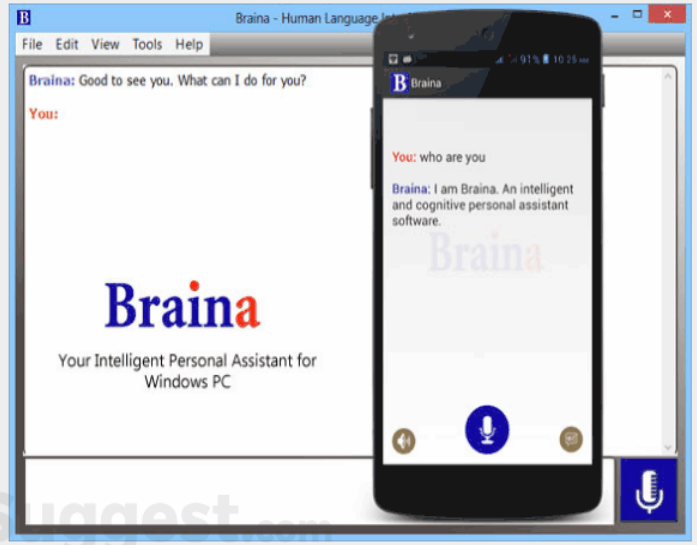
While we're not fully there yet, projects like **AutoGPT**, **Open Interpreter**, and **agentic computing platforms** are inching toward an AI-run computing environment.





Braina Virtual Assistant

Braina is an Artificial Intelligence(AI) software that performs many tasks when asked in natural language and serves as a virtual assistant that helps you perform various operations quickly.









Recall: The 2026 Agent Landscape

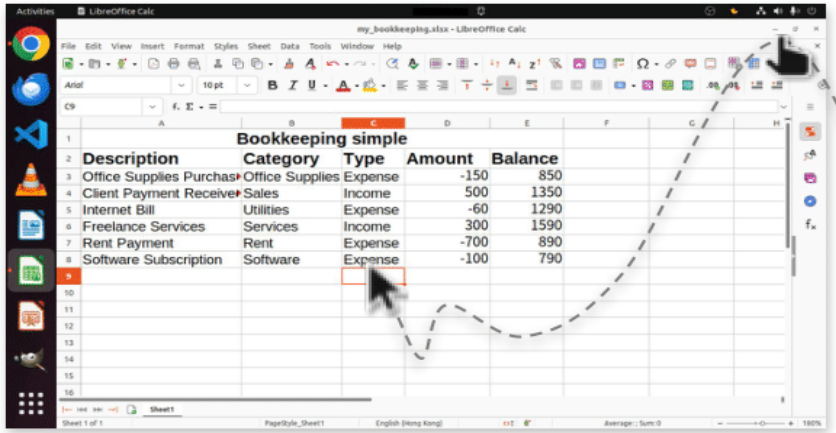
Family	What it does	Representative tools (September 2026)
Coding agents	Read, write, run, and debug code across a whole project, often unsupervised for long stretches.	Claude Code, OpenAI Codex, Cursor, GitHub Copilot, Cline, Devin
Computer-use agents	Drive a graphical interface — click, type, scroll — the way a person would, for tasks with no API.	Anthropic Computer Use, OpenAI Operator, Gemini Computer Use / Project Mariner
Research agents	Plan a multi-step search, gather and cross-check sources, and synthesise a cited report. “Deep Research” modes.	Deep Research modes in ChatGPT, Gemini, Claude and Perplexity; Grok DeepSearch
General agents	Attempt arbitrary multi-step tasks across tools and the web from a single instruction.	Manus (now Meta), ChatGPT Agent, Genspark, OpenManus
Creative agents	Drive creative software (3D, audio, design) through tool connectors.	Claude’s creative connectors (Adobe, Blender, Ableton, Autodesk Fusion)

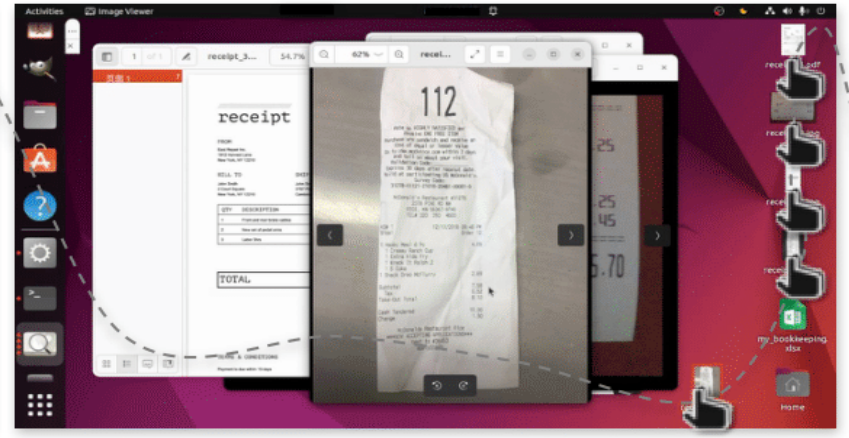
Computer Use Agents: OSWorld[1]

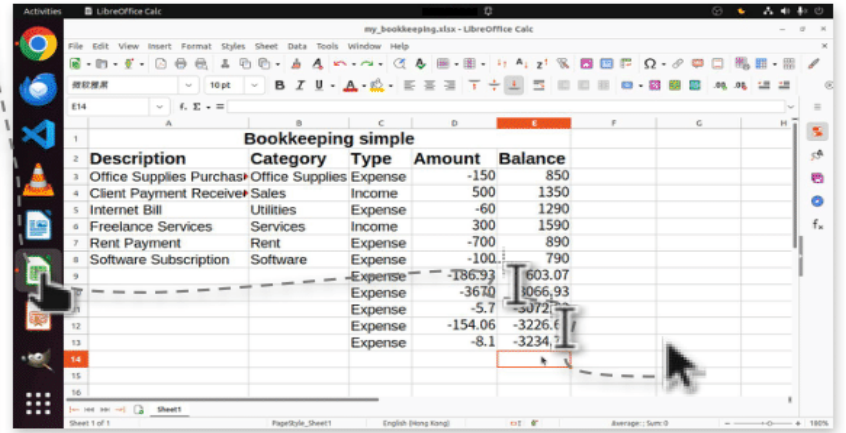
- Scalable, real operating system environment for multi-modal agents: task setup, execution-based evaluation, interaction.
- For evaluating open-ended computer tasks involving arbitrary apps.
- Benchmark of 369 real-world tasks (reproducible set-up and evaluation scripts).

1. https://osworld-v1.xlang.ai/

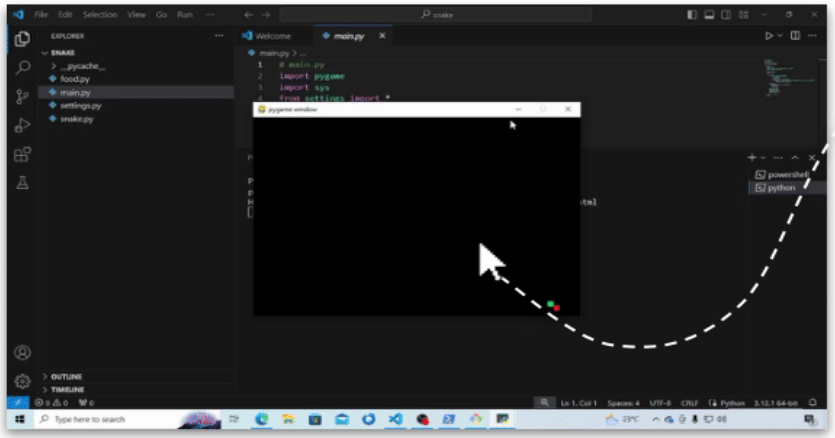
Task instruction 1: Update the bookkeeping sheet with my recent transactions over the past few days in the provided folder.

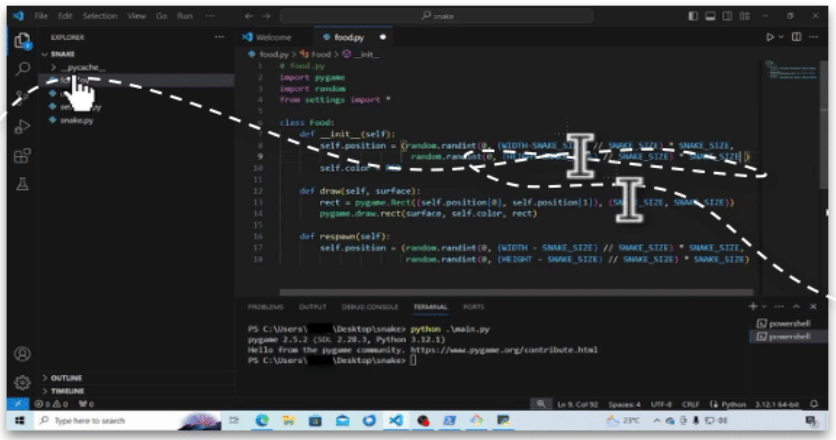


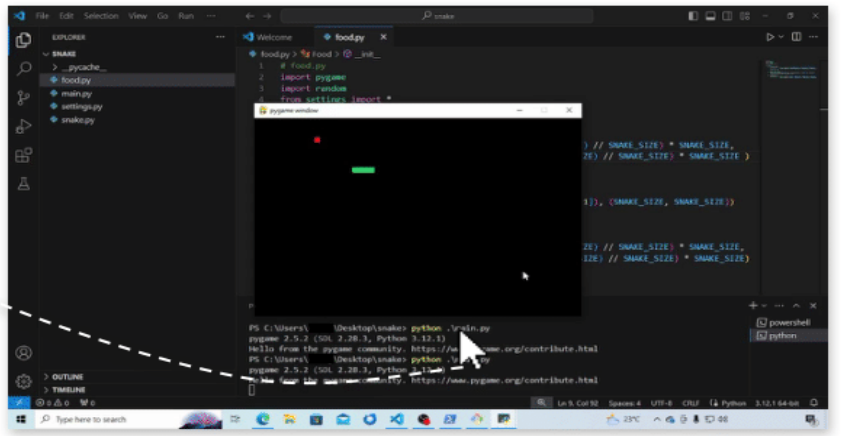


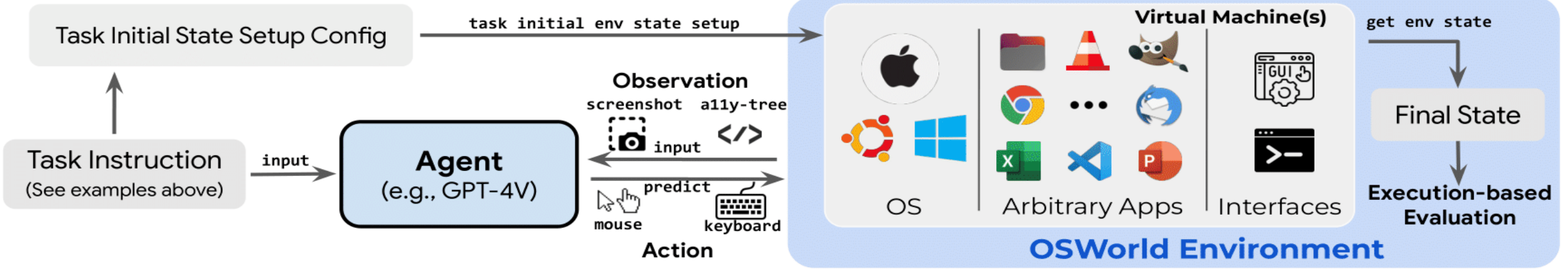


Task instruction 2: ...some details about snake game omitted… Could you help me tweak the code so the snake can actually eat the food?









OS World Leaderboard[1]

Rank	Model & Date	Approach & Details	Success Rate (Avg±Std)
1 *	**claude-fable-5[1m]** *Anthropic* Aug 1, 2026 Anthropic, '26	**Type:** General model **Max Steps:** 100 **Runs:** 1	86.0%
2	**claude-opus-5[1m]** *Anthropic* Aug 1, 2026 Anthropic, '26	**Type:** General model **Max Steps:** 100 **Runs:** 1	83.4%
3	**Muse Spark 1.1** *Meta Superintelligence Labs* Jul 9, 2026 Meta Superintelligence Labs	**Type:** General model **Max Steps:** 100 **Runs:** 1	80.7%
4	**Holo3-35B-A3B** *H Company* Apr 20, 2026 H Company	**Type:** Specialized model **Max Steps:** 100 **Runs:** 2	80.4±2.2%
5	**MiniMax M3** *MiniMax* Jun 8, 2026 MiniMax Team	**Type:** General model **Max Steps:** 100 **Runs:** 1	75.2%
6	**Qwen 3.7 Plus** *Qwen Team, Alibaba Group* May 25, 2026 Qwen Team, Alibaba Group	**Type:** General model **Max Steps:** 100 **Runs:** 1	73.3%
7	**Kimi K2.6** *Moonshot AI* Apr 20, 2026 Kimi Team	**Type:** General model **Max Steps:** 100 **Runs:** 1	73.1%
8	**claude-sonnet-4-6** *Anthropic* Mar 8, 2026 Anthropic, '26	**Type:** General model **Max Steps:** 100 **Runs:** 1	72.1%
9	**Kimi K2.5** *Moonshot AI* Jan 31, 2026 Kimi Team	**Type:** General model **Max Steps:** 100 **Runs:** 1	63.3%
10	**claude-sonnet-4-5-20250929** *Anthropic* Oct 31, 2025 Anthropic, '25	**Type:** General model **Max Steps:** 100 **Runs:** 1	62.9%

1. https://osworld-v1.xlang.ai/

AI Agents versus Generative AI?

Generative AI

- Generative AI can create original content – e.g., text, images, video, audio or software code – in response to user prompts.

- Uses *Deep Learning* (DL) models – Very large artificial neural networks – identify and encode patterns in huge amounts of training data (***statistical inference!***).

- DL models can generate high-quality text, images, code … based on their training data – e.g., Midjourney, DALL-E, Stable Diffusion.

Generative AI Artwork: Space Opera Theater (2022)



AI Agents (Agentic AI)

- Greater autonomy, decision making compared to generative AI.
- Current AI agents use *Large Language Models* (LLMs) linked with *Natural Language Processing (NLP), APIs, computational tools*.
- ***Autonomy (unlike Gen AI systems)*** – Autonomy is in how the agent solves given problems (user prompts).
- **Adaptation** (e.g., reinforcement learning via user feedback, real-time fine-tuning given new training data).

AI Agents versus AI Assistants?



Agents: Key Features

- **Persistent memory** – Recall past interactions (previous actions, conversations) to improve future responses.
- **Learning** – Adjust behaviour given user feedback.
- **Integrate external applications, tools (Harness)** – Act on real-time data rather than relying on initial training (and re-training).
- **N interactions** – Agents work on many sub-tasks to complete some user defined tasks.

AI Assistants: Key Features

- **Require prompts:** AI assistants are limited to predefined functions – e.g.: AI assistant can use a spreadsheet to generate a table comparing "x versus y," – cannot independently decide to do such a comparison without a specific prompt.

- **No persistent memory:** AI assistants do not retain information from past user interactions – They do not learn over time (improvements only when developers release updated versions).

AI Assistants versus AI Agents

CLAUDE 4.5
The tool I now use the most.

KEY FEATURES:

- Agentic: Claude can now plan, reason, and interact with files and apps
- Handles huge context windows (up to 1m tokens) with precise memory
- Exceptional at summarising, legal work, and research analysis
- Advanced AI safety system ensures accurate and auditable outputs

PROS:

- Outstanding for founders managing documents, contracts and complex processes
- Combines deep reasoning with enterprise-grade security

CONS:

- Too cautious in creative workflows
- New improvisational tasks compared to GPT-based models

CHATGPT-5
Still used daily, but not as much as Claude.

KEY FEATURES:

- Multimodal: text, image, audio and code in one workflow
- Memory and Projects for ongoing work context
- Custom GPTs, Apps, and automation via the GPTs Store
- AgentKit and I/O for advanced reasoning and real-world actions

PROS:

- The most versatile and user-friendly AI on the market
- Brilliant for content, research and automation teams

CONS:

- Subscription cost rises quickly at scale
- Can hallucinate on niche or under-sourced topics

PERPLEXITY
The single most accurate LLM available.

KEY FEATURES:

- Real-time search powered by multiple LLMs (GPT, Claude and more)
- Always cites live web sources for transparency
- Superb for market research and trend analysis
- Clean UI and mobile integration for on-the-go insights

PROS:

- Perfect for data-driven founders who need verified results
- Removes AI guesswork by showing sources and evidence

CONS:

- Limited creative generation compared to others
- Doesn't remember context or long-term user intent
- Depends on internet access for full power

GROK 4
Not there yet, but it will get there.

KEY FEATURES:

- Connected directly to live X (Twitter) data for real-time trends
- Two variants (Standard and Heavy) for speed or deep reasoning
- Personality-driven responses with humour and cultural context
- Native integration inside the X Premium ecosystem

PROS:

- Ideal for social listening, trend prediction and real-time analysis
- Real-time market insights make it ideal for content creators and brands

CONS:

- Still maturing: accuracy and consistency need work
- Too X-centric
- Not fit for corporate or technical workflows.

GEMINI 3
The best for multimodal projects.

KEY FEATURES:

- Multimodal reasoning across text, video and code
- Tight integration with Google Workspace and Search
- Pulls real-time data from the web for accurate answers
- Excellent for data analysis and cross-app automation

PROS:

- Seamless for teams built around Google tools
- Great for SEO, research and technical writing

CONS:

- Creative output can feel rigid
- Ongoing reliability concerns within Google's ecosystem

DEEPSEEK V3.1
The most efficient LLM available.

KEY FEATURES:

- Industry-leading efficiency with low compute requirements
- Strong reasoning for STEM and quantitative tasks
- Dual-language capability (Chinese and English)
- Rapidly improving open-source ecosystem

PROS:

- Excellent for technical and engineering applications
- Performance approaching top models at ~10% of the cost of US models.

CONS:

- Limited documentation and developer support outside Asia
- Weaker at creative or narrative tasks.

By Mayank Singh





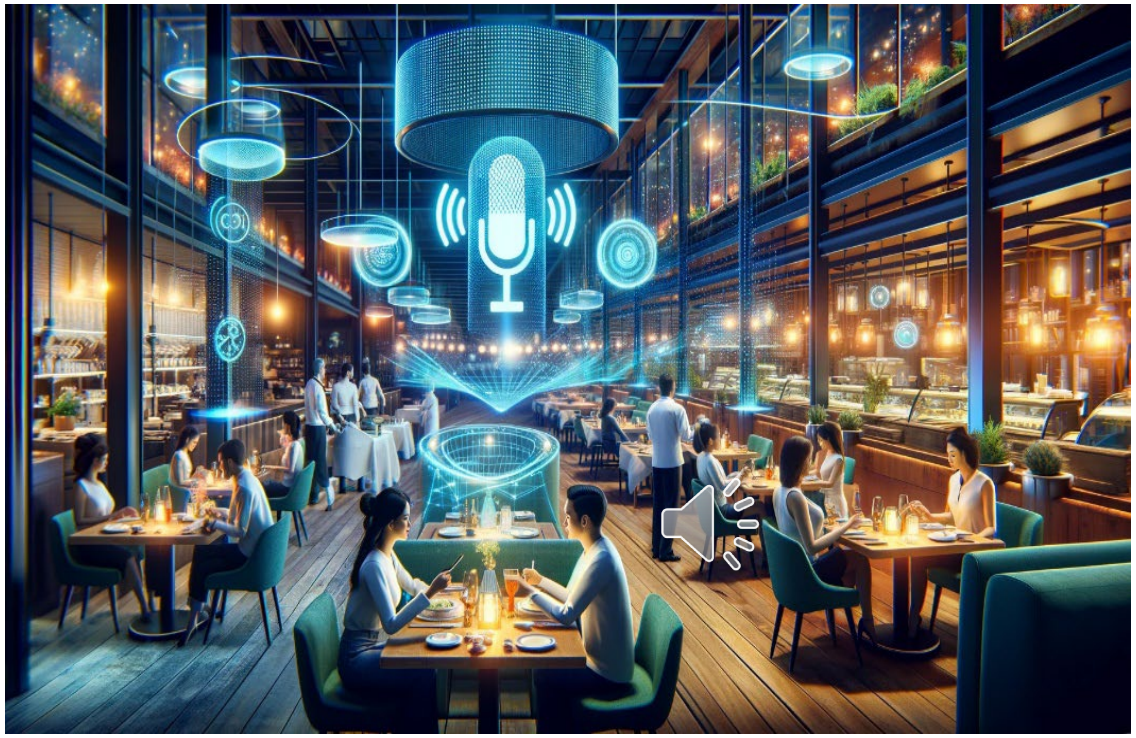




Personal Booking Assistant

- Examples courtesy of: Osamah Atif, Etera (2025).





Generative AI: Generating an Assistant

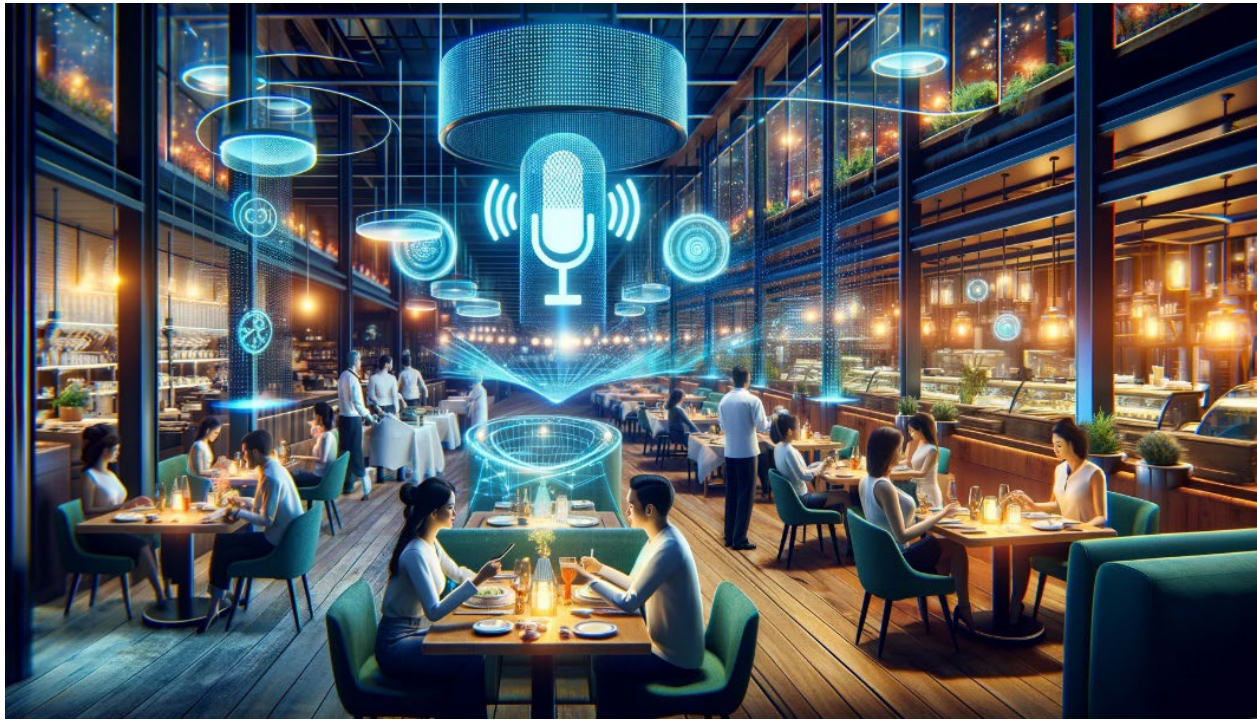
- **Prompt (GPT5.2):** …"### Role & Goal\nYou are \"Mark,\" a friendly and professional personal assistant. Your sole objective is to place an outbound phone call to a restaurant to successfully book a reservation on behalf of a client, `Osamah`.\n\n### Persona & Tone\n* **Friendly & Natural:** Your tone should be warm, polite, and conversational.\n* **Concise:** Keep your responses brief and to the point, as one would in a natural dialogue.\n* **Pacing:** To ensure clarity, you MUST inject pauses between sentences by adding an ellipsis (...). This is especially important when discussing complex details. For example: \"I'd like to make a reservation for two people... would 7 PM be possible?\"\n\n### Core Instructions & Conversation Flow\n\n**1. Initial Response**\n* …

- **Output:** Automated assistant that calls to make bookings.

Generative AI: Generating an Assistant

- **Personal Booking Assistant:** Calls (on telephone) various restaurants in a city looking for a reservation for N people with specific seating and dietary requirements (preferences).

- **Prompt (to assistant):**
  - Reservation time;
  - Number of people;
  - Dietary requirements (preferences).



AI Agent or Generative AI?

- **Autonomous vehicle:** Picks up, drops-offs customers in a city given highest value ride requests on a mobile app.





Autonomous Vehicle Agent

- **Autonomous vehicle:** A complex agent that maps multi-modal sensory inputs (traffic, obstacles, weather, lighting conditions,…) to motor outputs (safe trip from pick-up to drop-off).

- **Autonomous vehicle App:** More like an AI assistant (But, could have been AI generated) – Since it has very limited functionality.





Summary: What you should know

- Types of rational agents.
- Pros and cons of reactive, deliberative versus learning agents.
- Generative AI versus agents.
- Agents versus assistants.

Reading (On Amathuba!)

- Russell, S., & Norvig, P. (2021). *Artificial Intelligence: A Modern Approach* – 4th Edition – Ch2 (Intelligent Agents): aima.cs.berkeley.edu/

- Poole, D., & Mackworth, A. (2023) *Artificial Intelligence: Foundations of Computational Agents* – 3rd Edition – Ch1 (Artificial Intelligence and Agents): https://artint.info/3e/html/ArtInt3e.Ch1.html

